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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																														
B									B																																														
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-24 20kV Compressor Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																														
D									D																																														
E									E																																														
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="2">PT. ABB Sakti Industri</td><td rowspan="2">COVER SHEET SLD & GA Drawing MVD-24 20kV Compressor</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td></td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on: BE318042</td><td></td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-24 20kV Compressor	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.		Cont. 16	F
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A	SHEET INDEX								A					
B	PAGE		PAGE NAME			REVISION								
	==MVD-24=H00&GAD/1		COVER SHEET			0	1							
	==MVD-24=H00&GAD/2		INDEX OF SHEETS			0	1							
	==MVD-24=H00&GAD/3		GENERAL CHARACTERISTICS			0	1							
	==MVD-24=H00&GAD/4		REFERENCE DESIGNATIONS			0	1							
	==MVD-24=H00&GAD/5		REFERENCE DESIGNATIONS			0	1							
	==MVD-24=H00&GAD/6		REFERENCE DESIGNATIONS			0	1							
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	==MVD-24=H00&GAD/8		SINGLE LINE DIAGRAM			0	1							
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	==MVD-24=H00&GAD/11		FOUNDATION FRAMES			0	1							
	==MVD-24=H00&GAD/12		FOUNDATION FRAMES DETAILS			0	1							
	==MVD-24=H00&GAD/13		FOUNDATION FRAMES DETAILS			0	1							
	==MVD-24=H00&GAD/14		SECTION VIEW			0	1							
	C	==MVD-24=H00&GAD/15		SECTION VIEW			0	1						
==MVD-24=H00&GAD/16		SECTION VIEW			0	1								
D									D					
E									E					
F									F					
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-24 20kV Compressor		Scale		= H00
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0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri				Drawing Nbr.		Sh. 2
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A	CHARACTERISTICS									NOTES							A
	(IN COMPLIANCE WITH STANDARD IEC 62271-200)																
	SWITCHGEAR VERSION									= UNISEC							
	RATED VOLTAGE (Un)									= 24 kV							
B	OPERATING VOLTAGE									= 20 kV							
	RATED FREQUENCY (fr)									= 50 Hz							
	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)									= 125 kV							
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)									= 50 kV							
	RATED CURRENT OF MAIN BUSBARS (Ir)									= 630 A							
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)									= 20 kA							
C	RATED PEAK WITHSTAND CURRENT (Ip)									= 50 kA							
	RATED DURATION OF SHORT CIRCUIT (tk)									= 1 s							
	INTERNAL ARC CLASSIFICATION (IAC)									= AFLR / 16kAx1s							
	AMBIENT CONDITION									= NORMAL							
	AMBIENT AIR TEMPERATURE									= -5°C...+40°C							
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)									= IP3X							
D	DEGREE OF PROTECTION WITH OPEN DOORS									= IP2X							
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)									= 230VAC							
	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)									= 230VAC							
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)									= 230VAC							
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS									= 450/750V							
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS									= PVC							
E	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS									= 2.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS									= 1.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR MOTORS									= 2.5 mm²/BLACK							
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)									= 1.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS									= 1.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE									= 2.5 mm²/BLACK							
F	COMMUNICATION CABLE									= N/A							

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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS		A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR		
		-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR	
		-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.	
		-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT	
B	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE		B
					-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT		
		-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT	
		-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT	
		-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB3		HEATER LOCATED IN MOTOR CABINET	
		-BGK		POSITION SWITCH OF KEY LOCK		-EB4		HEATER FOR VENTILATION TEST	
		-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.	
		-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE	
		-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE	
		-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1	
		-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2	
		-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3	
		-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS	
		-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1	
C	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2		C
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3		
		-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER	
		-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS	
		-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1	
		-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2	
		-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3	
		-BM		HYGROSTAT		-FCT		THERMAL OVERLOAD RELEASES	
		-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS	
		-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR	
D	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR		D
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK		
		-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE	
		-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE	
		-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA3....-KFA9		AUXILIARY RELAYS OR CONTACTORS	
		-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFC		CLOSING RELAYS OR CONTACTORS	
		-BT		THERMOSTAT		-KFI		INTEGRATED CIRCUITS	
		-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY	
		-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFM		MICROPROCESSORS	
		-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFN		ANTIPUMPING RELAYS	
		-BU		BUCHHOLZ RELAY		-KFO		OPENING RELAYS OR CONTACTORS	
		-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)	
		-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFR		RECLOSING RELAYS	
		-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFS		SYNCHRONIZING DEVICES	
		-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS	
		-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFZ		CONTROLLERS	
		-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER	
						-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD	
						-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1	
						-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2	
						-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS	
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

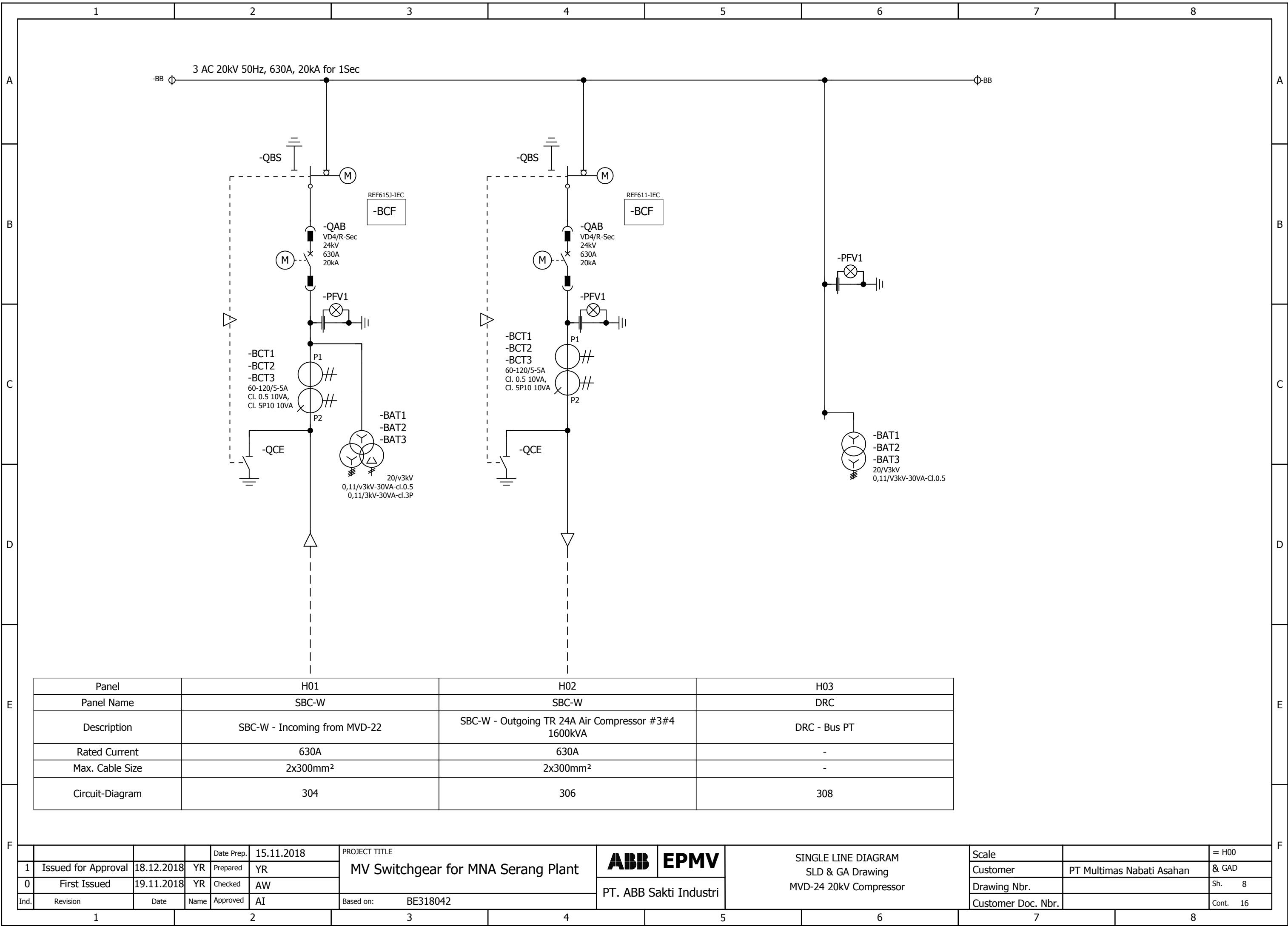
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	-TFA ACTIVE POWER TRANSDUCERS					-TFC CURRENT TRANSDUCERS					-TFF FREQUENCY TRANSDUCERS					-TFJ ACTIVE ENERGY TRANSDUCERS					-TFK REACTIVE ENERGY TRANSDUCERS					-TFM MULTIFUNCTION TRANSDUCERS					-TFP POWER-FACTOR TRANSDUCERS					-TFQ REACTIVE POWER TRANSDUCERS						-TFS SIGNAL CONVERTERS					-TFV VOLTAGE TRANSDUCERS					-WA BUSBARS					-WBC POWER CABLES					-WE EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.					-WGC CONTROL CABLES					-WGS COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)					-WH OPTICAL FIBRES					-XDA TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS					-XDA1 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1					-XDA2 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2					-XDA3 CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3					-XDA5 CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER					-XDB CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS					-XDB1 CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)					-XDB2 CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER					-XDB10...-XDB89 CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS					-XDB7 CONNECTOR FOR TRUCK POSITION CONTACTS					-XDB91 CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XDB92 CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XDB93 CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR					-XDB95 CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET					-XDC CUSTOMER TERMINAL BLOCK					-XDC1 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER					-XDC2 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR					-XDC3 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT					-XDC4 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS					-XDC5 CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER					-XDE TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE					-XDE1 TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE					-XDE2 TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2					-XDE3 TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2					-XDF TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS					-XDF1 CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR					-XDF2 CONNECTOR FOR REAR VENTILATOR CIRCUITS					-XDF3 CONNECTOR FOR FRONT VENTILATOR CIRCUITS					-XDH TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)					-XDI TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)					-XD11 TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS					-XD12 TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING					-XD13 TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION					-XD14 TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS					-XD16 TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS					-XD17 TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS					-XD18 TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS					-XD19 TERMINAL BLOCK FOR SPECIAL USE					-XDJ TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD					-XDJ1 TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS					-XDM SEALABLE TERMINAL BLOCK FOR MEASUREMENT					-XDS CURRENT SOCKETS					-XDT TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB					-XDU CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					-XDU1 TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					-XDV TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS					-XDV1 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1					-XDV2 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2					-XDV3 CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3					-XDV4 CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR					-XDX SUPPORT TERMINAL BLOCK					-XE1 EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE					-XE2 EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE					-XE5 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE					-XE6 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE					-XE7 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE					-XE8 EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE					-XE9 EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT					F																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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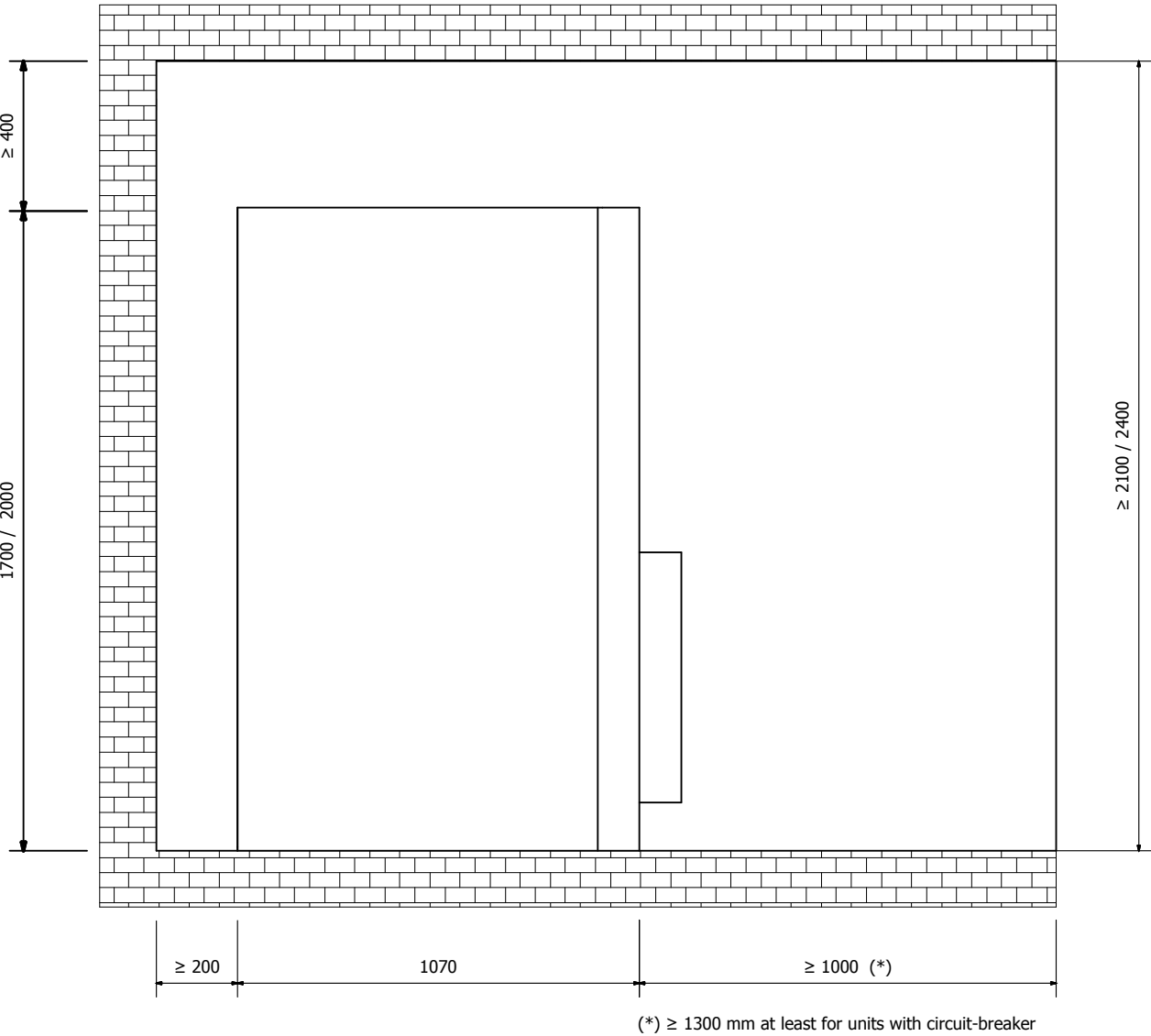
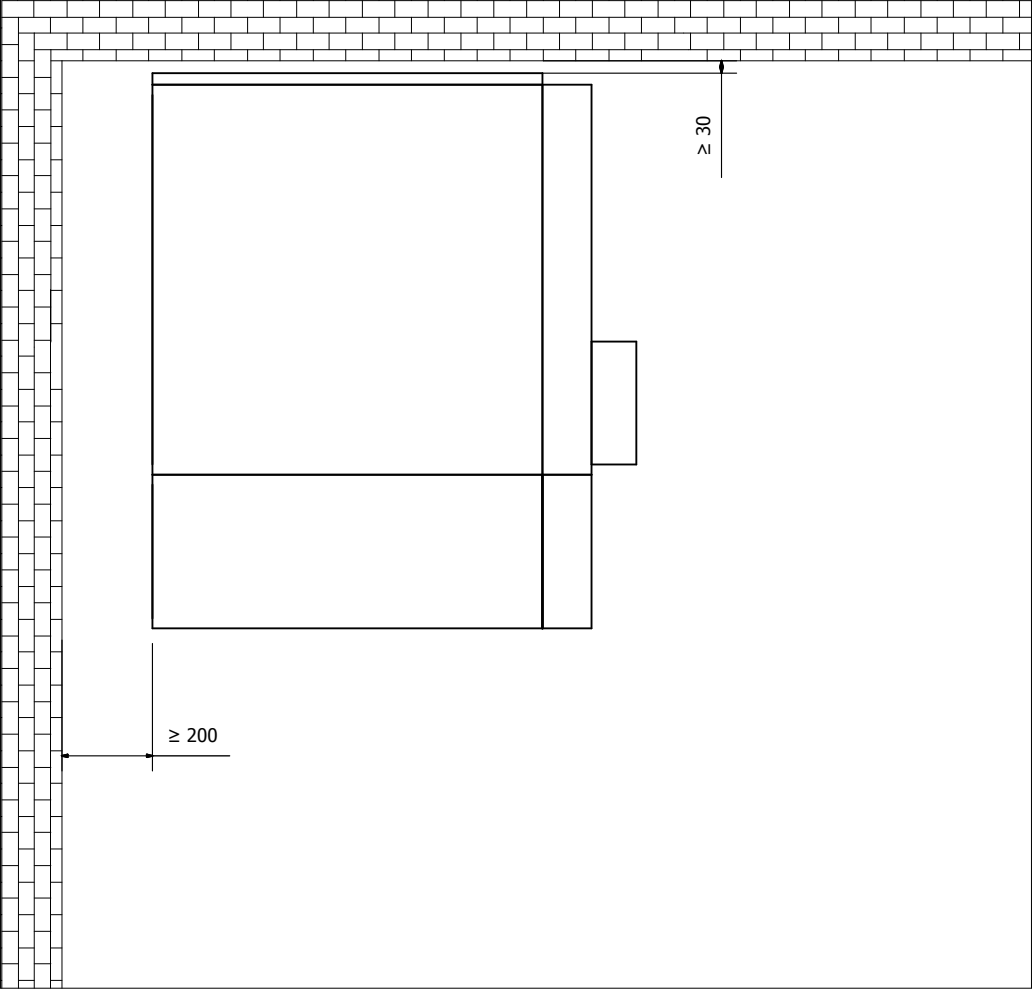
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RULES FOR INSTALLATION IN THE ROOM

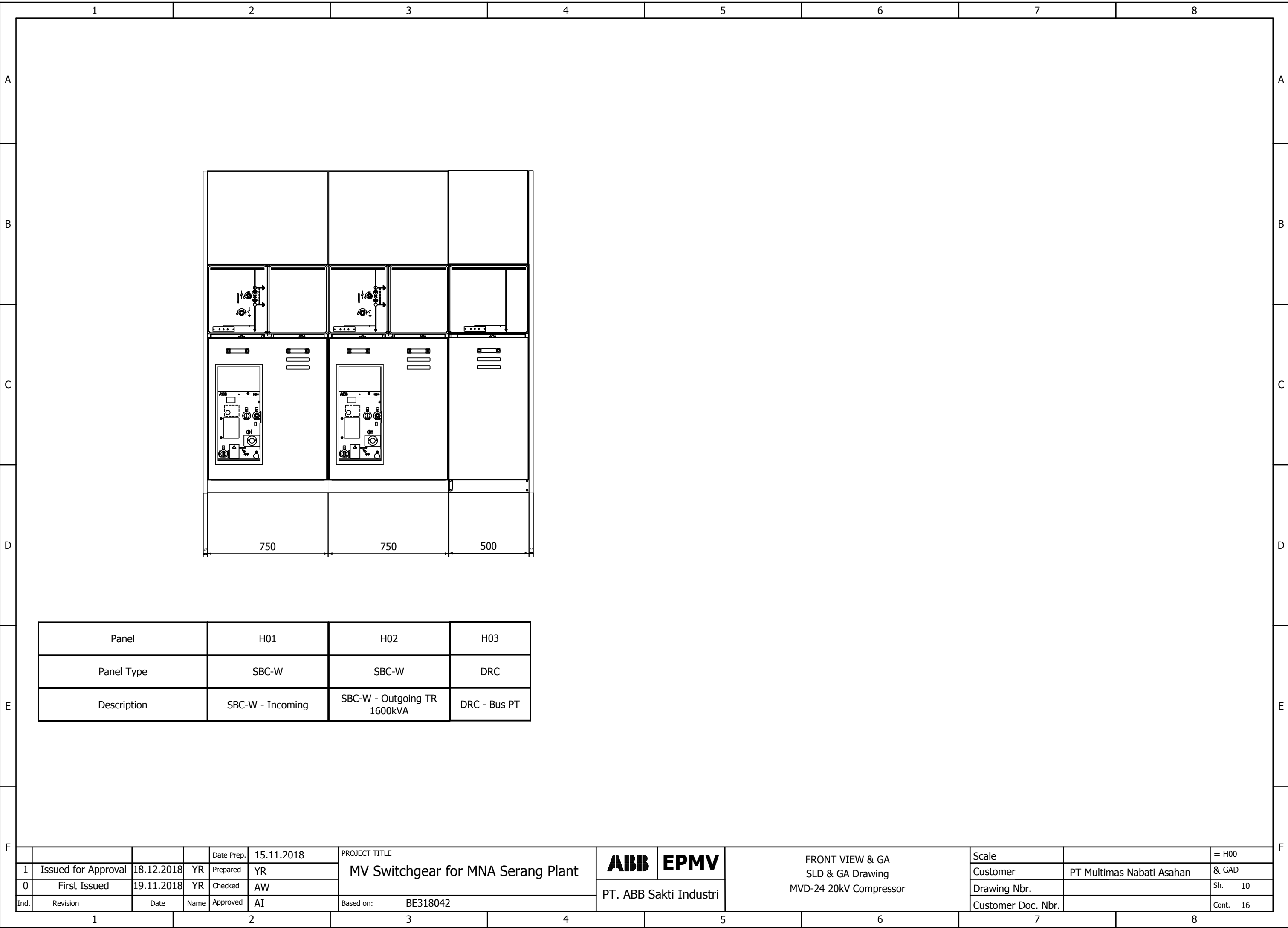
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

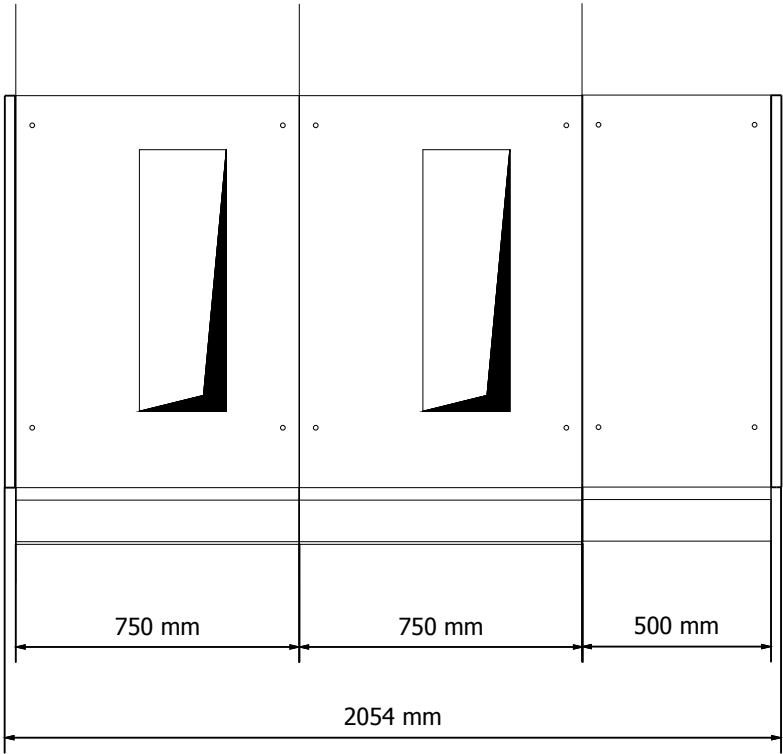
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			INSTALLATION RULES SLD & GA Drawing MVD-24 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.			Sh. 9	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 16	

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FOUNDATION FRAMES

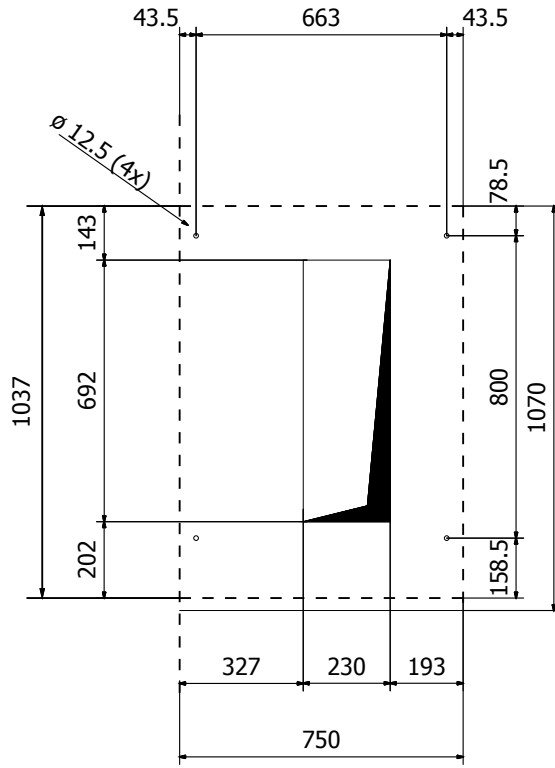
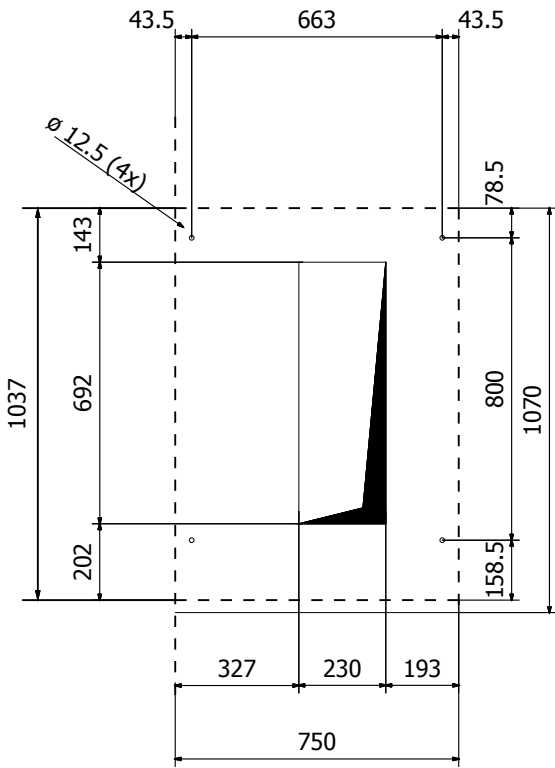


Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-24 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

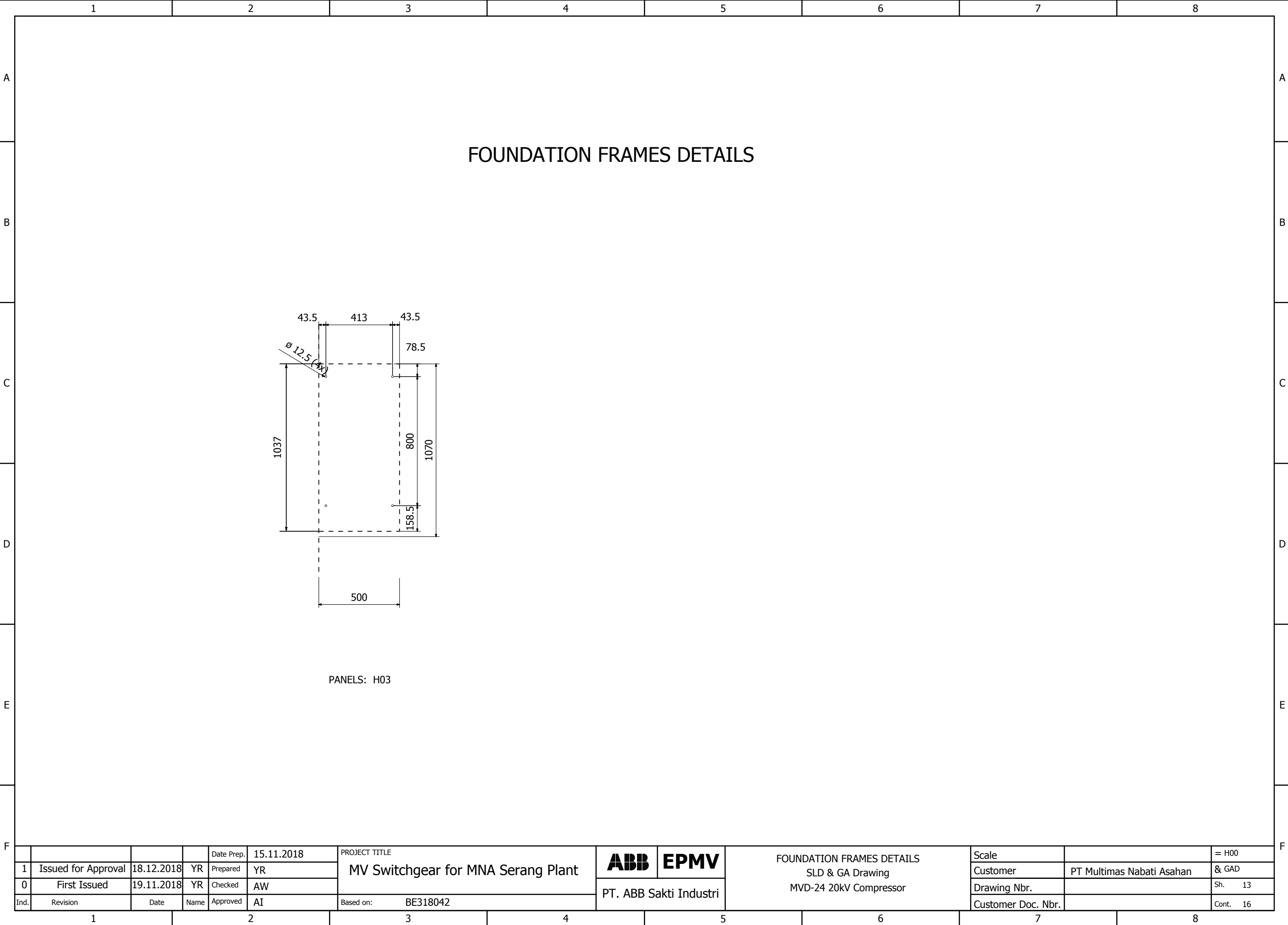
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FOUNDATION FRAMES DETAILS

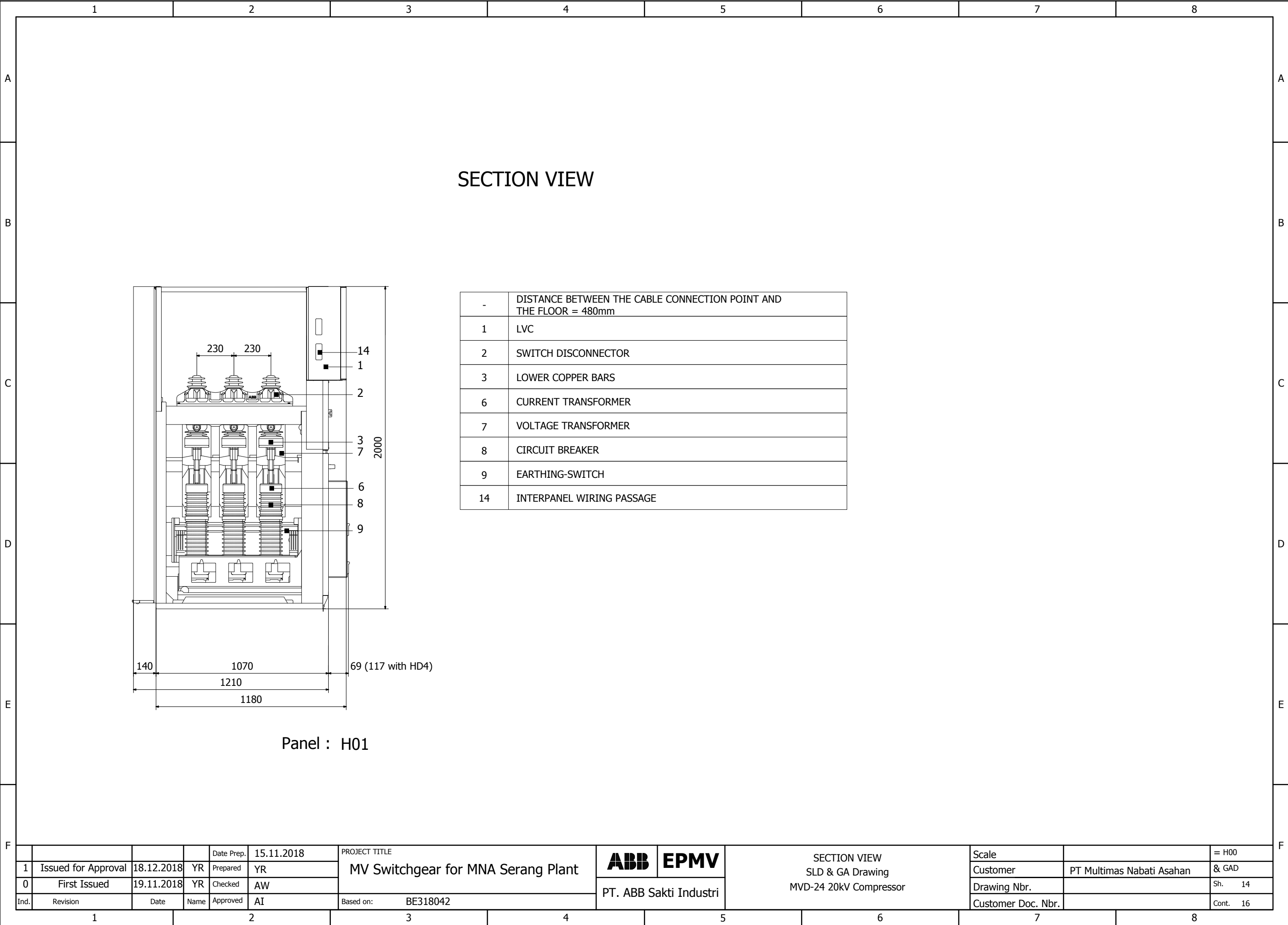


Ind.				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-24 20kV Compressor	Scale		= H00
	1	Issued for Approval	18.12.2018	YR	Prepared	YR			Customer	PT Multimas Nabati Asahan	& GAD
	0	First Issued	19.11.2018	YR	Checked	AW			Drawing Nbr.		Sh. 12
		Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.		Cont. 16

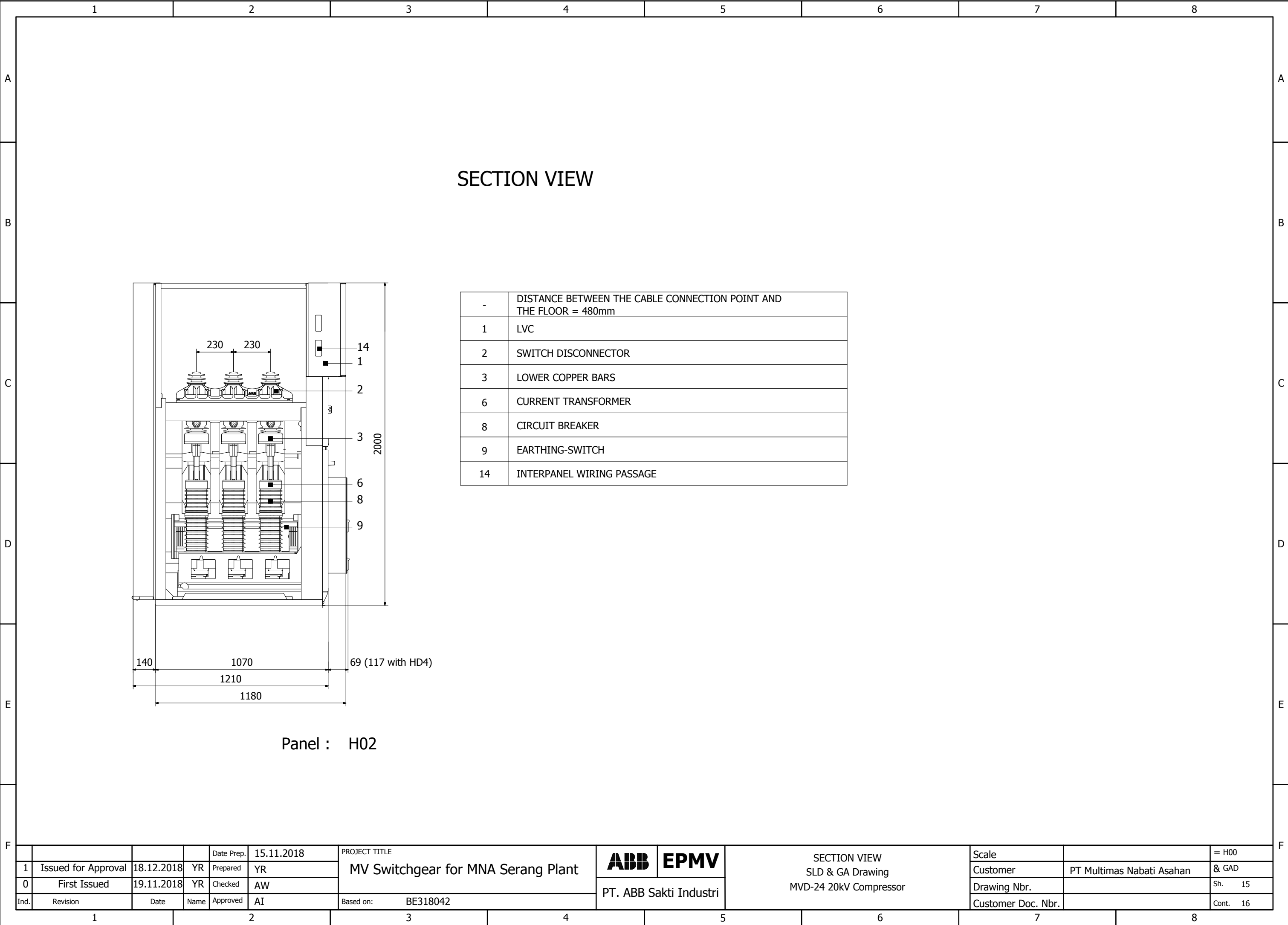
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